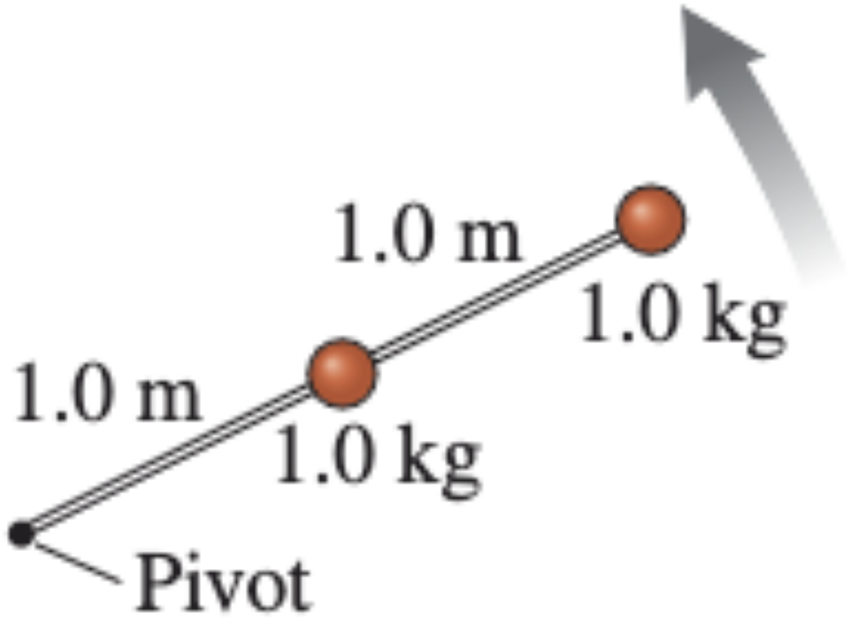
Problem session 10/20

P. 226 - 33

- A $250\ \mathrm{g}$ ball is launched with a speed of $35\ \mathrm{m/s}$ at a 30° angle. A strong headwind exerts a constant horizontal drag force on the ball. What is the magnitude of the drag force if the wind reduces the ball's travel distance by 20%?

P. 227

- The figure shows two small $1\,\mathrm{kg}$ masses connected by massless but rigid $1.0\,\mathrm{m}$ long rods (two of them). What is the tension in the rod that connects the pivot if the masses rotate at 30 rpm in a horizontal circle?



- The attic and kinetic friction coefficients of the surface in the figure are 0.90 and 0.50 respectively. The hanging mass is increased to the point where the two stacked masses are on the verge of sliding. If the top mass m is then removed, what is the acceleration of mass $2m$? The rope is massless, and the pulley is massless and frictionless

